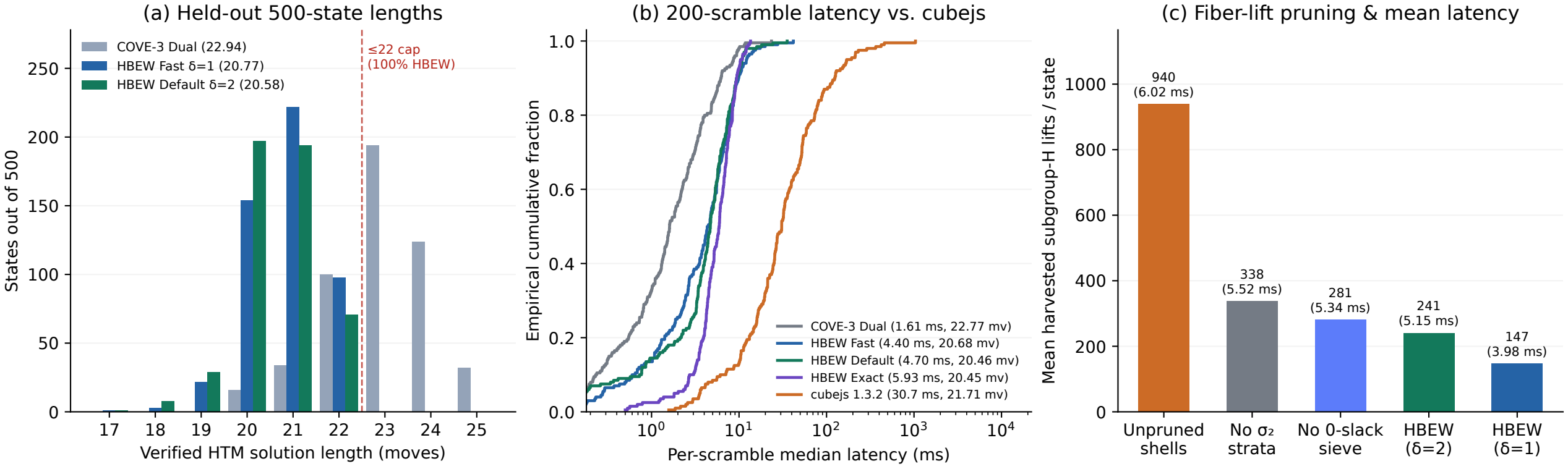
(a) Held-out 500-state lengths
COVE-3 Dual (22.94)
HBEW Fast δ=1 (20.77)
HBEW Default δ=2 (20.58)
≤22 cap
(100% HBEW)
States out of 500
Verified HTM solution length (moves)
(b) 200-scramble latency vs. cubejs
Empirical cumulative fraction
Per-scramble median latency (ms)
COVE-3 Dual (1.61 ms, 22.77 mv)
HBEW Fast (4.40 ms, 20.68 mv)
HBEW Default (4.70 ms, 20.46 mv)
HBEW Exact (5.93 ms, 20.45 mv)
cubejs 1.3.2 (30.7 ms, 21.71 mv)
(c) Fiber-lift pruning & mean latency
Mean harvested subgroup-H lifts / state
940
(6.02 ms)
338
(5.52 ms)
281
(5.34 ms)
241
(5.15 ms)
147
(3.98 ms)
Unpruned shells
No σ₂ strata
No 0-slack sieve
HBEW (δ=2)
HBEW (δ=1)